

PC LAB 7

IT Number – IT24101472

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Exercise 1

```
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB...  
/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise01  
manuth@manuth-LOQ-15IRH8: ~/Documents/mpi manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Vc x  
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7$ cd Exerc  
ise01  
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0  
1$ nano sum_bcast.c  
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0  
1$ mpicc sum_bcast.c -o sum_bcast  
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0  
1$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_bcast  
Root filled array with values 1 to 1000000  
Rank 0: summed indices [0, 250000) => local_sum = 31250125000  
Rank 1: summed indices [250000, 500000) => local_sum = 93750125000  
Rank 3: summed indices [750000, 1000000) => local_sum = 218750125000  
Rank 2: summed indices [500000, 750000) => local_sum = 156250125000  
  
[Bcast] Total sum = 500000500000  
[Bcast] Expected = 500000500000  
[Bcast] Correct? = YES  
[Bcast] Time = 0.0342 sec  
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0  
1$
```

Exercise 2

```
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB...
/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise02

manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
2$ ls
exercise02.txt  sum_scatter.c
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
2$ mpicc sum_scatter.c -o sum_scatter
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
2$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_scatter
Root filled array with values 1 to 1000000
Rank 2: local_sum = 156250125000
Rank 1: local_sum = 93750125000
Rank 3: local_sum = 218750125000
Rank 0: local_sum = 31250125000

[Scatter] Total sum = 500000500000
[Scatter] Expected  = 500000500000
[Scatter] Correct?   = YES
[Scatter] Time       = 0.0011 sec
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
2$ █
```

Exercise 3

```
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
3$ nano sum_gather.c
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
3$ mpicc sum_gather.c -o sum_gather
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
3$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_gather
Root filled array with values 1 to 1000000
Rank 1: local_sum = 93750125000
Rank 2: local_sum = 156250125000
Rank 0: local_sum = 31250125000
Rank 3: local_sum = 218750125000

[Gather] Total sum = 500000500000
[Gather] Expected  = 500000500000
[Gather] Correct?   = YES
[Gather] Time       = 0.0010 sec
manuth@manuth-LOQ-15IRH8: /run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
3$ █
```

Exercise 4

```
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
4$ mpicc sum_reduce.c -o sum_reduce
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
4$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_reduce
Root filled array with values 1 to 1000000
Rank 0: local_sum = 31250125000
Rank 1: local_sum = 93750125000
Rank 3: local_sum = 218750125000
Rank 2: local_sum = 156250125000

[Reduce] Total sum = 500000500000
[Reduce] Expected  = 500000500000
[Reduce] Correct?   = YES
[Reduce] Time       = 0.0012 sec
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
4$ █
```

Exercise 5

```
6$ nano sum_allreduce.c
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$ mpicc sum_allreduce.c -o sum_allreduce
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_allreduce
Root filled array with values 1 to 1000000
Rank 0: local_sum = 31250125000, total_sum = 500000500000, contribution = 6.25%

[Allreduce] Total sum = 500000500000
[Allreduce] Expected  = 500000500000
[Allreduce] Correct?   = YES
[Allreduce] Time       = 0.0017 sec
Rank 2: local_sum = 156250125000, total_sum = 500000500000, contribution = 31.25%
Rank 1: local_sum = 93750125000, total_sum = 500000500000, contribution = 18.75%
Rank 3: local_sum = 218750125000, total_sum = 500000500000, contribution = 43.75%
manuth@manuth-LOQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$
```

Exercise 6

```
manuth@manuth-LQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$ nano sum_scan.c
manuth@manuth-LQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$ mpicc -o sum_scan sum_scan.c
manuth@manuth-LQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$ OMPI_MCA_btl=self,sm mpirun -np 4 ./sum_scan
Root filled array with values 1 to 1000000
Rank 0: local_sum = 31250125000, prefix_sum = 31250125000, sum_before_me = 0
Rank 1: local_sum = 93750125000, prefix_sum = 125000250000, sum_before_me = 31250125000
Rank 2: local_sum = 156250125000, prefix_sum = 281250375000, sum_before_me = 125000250000
Rank 3: local_sum = 218750125000, prefix_sum = 500000500000, sum_before_me = 281250375000

[Scan] Final prefix sum = 500000500000
[Scan] Expected      = 500000500000
[Scan] Correct?      = YES
manuth@manuth-LQ-15IRH8:/run/media/manuth/New Volume1/sliit/Y3S1/new/PC/LAB/PC_LAB_7/Exercise0
6$
```

Exercise 7

Program	Collective(s) Used	Memory Allocation	Manual Sum on Root?	Final Result Available Where?
sum_bcast.c	MPI_Bcast + MPI_Send/Recv	Every process has full array	Yes	Root only
sum_scatter.c	MPI_Scatter + MPI_Send/Recv	Root full array, others only chunk	Yes	Root only
sum_gather.c	MPI_Scatter + MPI_Gather	Root full array, others only chunk	Yes	Root only
sum_reduce.c	MPI_Scatter + MPI_Reduce	Root full array, others only chunk	No	Root only
sum_allreduce.c	MPI_Scatter + MPI_Allreduce	Root full array, others only chunk	No	All processes
sum_scan.c	MPI_Scatter + MPI_Scan	Root full array, others only chunk	No	Different prefix result on each rank

Timing table

Program	2 Processes	4 Processes	8 Processes
MPI_Bcast	0.0017 s	0.0040 s	0.0055 s
MPI_Scatter	0.0020 s	0.0011 s	0.0013 s
MPI_Gather	0.0018 s	0.0011 s	0.0016 s
MPI_Reduce	0.0018 s	0.0012 s	0.0016 s
MPI_Allreduce	0.0019 s	0.0011 s	0.0009 s
MPI_Scan	0.0047 s	0.0043 s	0.0052 s